Hong-Son Nguyen

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Research interests

Human-aware and social robot navigation; multi-agent pedestrian trajectory prediction; reinforcement learning for planning and control; human–robot interaction and pedestrian comfort. I am interested in representations that encode human state, social interaction structure and group behaviour, and in validating them on physical robots rather than in simulation alone.

Education

2024 – present

M.S. in Robotics and Control Systems

Department of Mechanical Engineering, National Cheng Kung University, Tainan, Taiwan

Advisor: Prof. Yen-Chen Liu. GPA 3.52/4.0 (87.89/100).

2019 – 2023

B.S. in Mechatronic Engineering

Hanoi University of Industry, Hanoi, Vietnam

Advisor: Assoc. Prof. Anh-Tu Nguyen. GPA 3.05/4.0.
Thesis: *Development of an Autonomous Hexapod Robot using Deep Reinforcement Learning Methods.* Graded 94 and selected as best thesis.

Research experience

Feb 2024 – present

Research Fellow

Networked Robotic Systems Laboratory, National Cheng Kung University

- Built an end-to-end human-aware navigation pipeline covering sensing, human-state representation, prediction, planning, learning and control, validated on an Agilex Scout Mini with ZED-X stereo vision, Velodyne VLP-16 LiDAR and Jetson AGX Orin compute.
- Developed Social Graph Mamba, an attention-free trajectory forecasting model using selective state space models over dynamic interaction graphs, reaching linear rather than quadratic complexity in the number of agents.
- Ran pedestrian studies with human participants, including comfort surveys and their statistical analysis, feeding empirical comfort models back into the navigation stack.

Funded projects: *Advanced Resilient and Intelligent Control for Heterogeneous Cyber-Physical Mobile Robot Systems*, MOST, Taiwan (Mar – Jul 2024); *Designing and Implementing Generative AI Service Robots: Exploring Applications in Companionship and Home Services*, NSTC, Taiwan (Aug 2024 – Aug 2025); *Trust-Integrated Intelligent Control for Socially Aware Robot Navigation in Dynamic and Crowded Environments*, NSTC, Taiwan (Aug 2025 – present).

2022 – Jan 2024

Research Assistant and Team Leader

Hanoi Intelligent Robotics Laboratory, Hanoi University of Industry

- Led a student research team on reinforcement learning for autonomous mobile robots, producing the group's first published navigation work.
- Implemented and benchmarked Q-Learning and SARSA navigation policies in Gazebo, with experimental validation on physical platforms.

Funded project: *Development of the Localization System for Mobile Robots Working in the Outdoor Environment*, Hanoi University of Industry (HAUI), Vietnam (Dec 2022 – Dec 2023).

Publications

Social Graph Mamba: Forecasting Pedestrian Movements Based on Social Context
Hong-Son Nguyen, Yen-Chen Liu
arXiv preprint arXiv:2608.21411, 2026 arXiv

Stability and Comfort in Mobile Robot–Pedestrian Interactions
Alireza Jafari, **Hong-Son Nguyen**, Yen-Chen Liu
arXiv preprint arXiv:2607.17604, 2026 arXiv

Empirical Prediction of Pedestrian Comfort in Mobile Robot–Pedestrian Encounters
Alireza Jafari, **Hong-Son Nguyen**, Yen-Chen Liu
IEEE International Conference on Robotics and Automation (ICRA), 2026 arXiv

Combination of Deep Reinforcement Learning and Pure-Pursuit Algorithm for Adaptive Autonomous Mobile Robot Navigation
Hong-Son Nguyen, Anh-Tu Nguyen
International Conference on Intelligent Systems and Networks (ICISN), Lecture Notes in Networks and Systems, vol. 1077, pp. 563–572, Springer, 2024 Springer

The Application of the Reinforcement Learning Method for Mobile Robot Navigation in an Unknown Environment
Anh-Tu Nguyen, **Hong-Son Nguyen**
International Conference on Material, Machines and Methods for Sustainable Development (MMMS), Lecture Notes in Mechanical Engineering, pp. 201–207, Springer, 2024 Springer

Technical skills

Languages	Python, C++, MATLAB
Machine Learning	PyTorch, deep reinforcement learning, Attention mechanism, selective state space models, Sim2Sim and Sim2Real transfer
Libraries	ROS 2, NVIDIA Isaac Sim / IsaacLab, Gazebo, MuJoCo, CUDA
Platforms	Agilex Scout Mini, Unitree Go2, Jetson AGX Orin

Awards and service

2024 – 2026 **Distinguished International Student Scholarship**
National Cheng Kung University

2023 **Scholarship for excellent students**
Department of Mechanical Engineering, Hanoi University of Industry

2022 **First prize, Students Scientific Research Competition**
Hanoi University of Industry

Since 2022 **Student Member**
American Society of Mechanical Engineers (ASME)

Languages

English	IELTS 6.0